

SUMMARY

AI/ML Engineer building production ML and LLM systems with Python, PyTorch, FastAPI, Docker, and CI/CD. Experienced in taking projects from data preparation and model development through evaluation and deployment, with a focus on privacy-preserving ML, synthetic data, and RAG applications.

TECHNICAL SKILLS

**Programming:** Python, SQL, C/C++, Java, JavaScript, Bash  
**ML & AI:** PyTorch, Scikit-learn, Pandas, NumPy, OpenCV  
**GenAI & LLMs:** RAG, Vector DBs, Prompt Engineering  
**Databases:** MySQL, PostgreSQL, SQLite, MongoDB  
**DevOps & Tools:** Docker, FastAPI, REST APIs, Git, CI/CD

PROFESSIONAL EXPERIENCE

- AI Engineer | MyVellum — Remote** Dec. 2025 – Present
- Built and containerized a FastAPI RAG service with Groq-hosted LLMs, vector-backed retrieval, and modular prompt orchestration
  - Implemented CI/CD evaluation gates that block deployments when retrieval-quality metrics fall below defined thresholds
- Researcher & Teaching Assistant | Eötvös Loránd University — Budapest, Hungary** Sep. 2025 – Present
- Built and optimized DP-SGD federated-learning pipelines for privacy-preserving synthetic data generation ( $\epsilon = 1$ )
  - Developed reproducible data preparation, feature engineering, and evaluation workflows for privacy-utility benchmarking
- Software Engineering Fellow | ALX Africa — Remote** Jun. 2023 – Feb. 2025
- Designed and implemented backend services and RESTful APIs with database integration in Linux environments
  - Built modular, maintainable software solutions through iterative development and code reviews

PROJECTS

- CASPER**  2026
- Python Differential Privacy RAG T5 MedCPT Docker
- Built a privacy-preserving RAG framework that clusters sensitive medical records with DP-KMeans and inverts noisy centroids into synthetic prototypes, enabling LLMs to retrieve on private data without accessing raw records
  - Evaluated the privacy-utility trade-off, achieving Recall@10 = 0.91 and MIA AUC  $\approx$  0.50 (random-guessing)
- Context-Aware Access Control for IoT Devices**  2026
- FastAPI Python Multi-Agent LLMs Docker Wazuh Prometheus
- Built a hybrid zero-trust IoT authorization framework combining a deterministic validation pre-filter ( $< 1$  ms fast path) with multi-agent LLM reasoning (97 ms slow path) for autonomous context-aware access control
  - Achieved 94.1% decision accuracy vs. 82.4% for the RBAC baseline and blocked 100% of 105 red-team adversarial attacks, with fail-secure DENY defaults on LLM rate limits or timeouts
- EVA — École Virtuelle Algérienne**  2023 – Present
- Next.js React Tailwind CSS EN/AR i18n Sentry Vercel
- Built and deployed a production e-learning platform serving Algerian students across primary, middle (BEM), and high school (BAC), aligned with the official national curriculum
  - Implemented bilingual EN/AR interfaces with full RTL layout, course search, learning paths, instructor profiles, and voucher-code offline payments; integrated Sentry for production error monitoring

PUBLICATIONS

- CASPER: Contextual Anonymized Synthetic Prototypes for Enhanced Retrieval in Medical Information Systems** 2026  
Ahmed Fouad Lagha, Loubna Seddiki, Oumaima Araar, Imre Lendák — *IDEAL 2026, Main Track*
- Context-Aware Access Control for IoT Devices Using Two-Role LLM Reasoning: A Security Evaluation** 2026  
Ahmed Fouad Lagha, Loubna Seddiki, Oumaima Araar, Imre Lendák — *ITAT 2026, Workshop on Applied Security (WAS)*
- A Survey of IoMT Datasets: Availability, Applications, and Research Insights** 2026  
Loubna Seddiki, Ahmed Fouad Lagha, Imre Lendák — *IEEE ICC 2026*

EDUCATION

- PhD in Informatics | Eötvös Loránd University, Hungary** Sep. 2025 – Expected 2029
- M.Sc. in Artificial Intelligence | University of Khenchela, Algeria** Sep. 2023 – Jun. 2025
- B.Sc. in Computer Science | University of Khenchela, Algeria** Sep. 2020 – Jun. 2023

CERTIFICATIONS & AWARDS

- Stipendium Hungaricum Scholarship:** Fully funded PhD studies at ELTE Sep. 2025
- 2nd Place — Starthon'25 Hackathon:** University of El Oued, Algeria Apr. 2025
- Software Engineering (Back-end Specialization):** ALX Africa  Feb. 2025
- AI Career Essentials:** ALX AiCE  Jul. 2024

LANGUAGES

Arabic (Native), Tamazight (Native), French (Fluent), English (Fluent)